

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – AI Powered Interview Trainer Agent (RAG Based)

Domain of Project – Education

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Problem Statement – Interview Trainer Agent

The Challenge

Cracking a job interview requires more than technical knowledge - candidates often struggle with structuring answers, anticipating role-specific questions, and presenting their experience with confidence. Information about interview expectations, behavioral frameworks, and industry-specific question patterns is scattered across unreliable sources. Without personalized, real-time guidance, many candidates fail to prepare effectively or perform below their true potential in competitive hiring environments.

The Objective

Build an AI-powered Interview Trainer Agent that delivers personalized interview preparation and coaching for job seekers across experience levels and roles. The solution should:

Personalized Question Sets :

Generate tailored interview questions based on profile name, experience level, job role, and resume/JD.

Interview Data RAG Retrieval :

Fetch and summarize role-specific questions, industry expectations, and HR guidelines from recruitment portals, professional networks, and company interview databases.

Response Practice & Feedback :

Allow users to submit answers via text, resume upload, or voice, and provide instant evaluation with model answers and improvement tips.

Comprehensive Skill Focus :

Assess both technical competency and soft skills, including behavioral scenarios using frameworks like STAR.

Multi-Agent System :

Question Retrieval Agent (fetches role-specific interview questions & guidelines)

Answer Recommendation Agent (generates model answers & personalized responses)

Interview Advisory Agent (behavioral, HR, and soft-skill coaching)

Response Log & Feedback Agent (tracks practice attempts + provides improvement analysis)

Progress Dashboard :

Display a personalized prep kit overview with technical questions, behavioral/HR STAR-format questions, targeted improvement tips, and a confidence checklist, with an option to download the full kit for continued practice.

Proposed Solution – AI Powered Interview Trainer Agent with RAG

The proposed solution is an AI-powered Interview Trainer Agent built using IBM watsonx.ai and Granite Models to generate personalized interview questions based on a candidate's profile. The system uses a Retrieval-Augmented Generation (RAG) pipeline, where a curated knowledge base of text files containing interview questions, industry expectations, and HR guidelines is chunked, embedded, and stored in a vector database.

The workflow is as follows:

User Input :

The user provides their job title, experience level, and uploads a resume or job description.

RAG Retrieval :

The system retrieves the most relevant chunks from the knowledge base (text files) based on the role and experience level provided.

Question Generation :

IBM Granite processes the retrieved context along with the resume/JD to generate a tailored set of interview questions, covering both technical and behavioral/HR aspects.

Output :

The user receives a personalized set of interview questions relevant to their target role and background, helping them prepare more effectively.

The solution includes a user-friendly dashboard (built with Streamlit) that visualizes the personalized prep kit, highlights technical and behavioral/HR questions, provides model answers and improvement tips, and tracks a confidence checklist to help candidates monitor their interview readiness. IBM Bob is used to accelerate development of the underlying agent and tool logic, while watsonx.ai ensures reliable model deployment, embeddings, and retrieval from the RAG knowledge base. Granite models power reasoning, question generation, and personalization based on the candidate's resume, job description, and experience level. Overall, this solution delivers an end-to-end intelligent interview preparation assistant that builds candidate confidence, sharpens responses, and increases success rates through personalized, RAG-grounded interview questions and real-time preparation support.

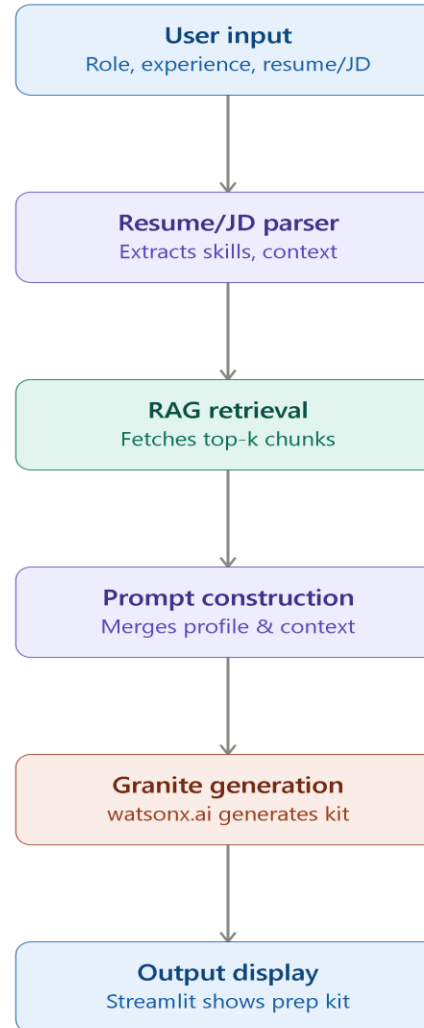
Technology Used

1. **IBM Bob** – An AI-powered IDE with watsonx.ai integrated, used to accelerate development of the Interview Trainer Agent's code, tools, and agent logic
2. **IBM Granite Model** – e.g., ibm-granite-3-3-8b-instruct – For natural language understanding, reasoning, and personalized question/answer generation
3. **IBM Cloud** – Provides the infrastructure to build, deploy, and scale the Interview Trainer Agent efficiently. IBM Cloud enables a reliable, secure, and scalable environment for deploying the agent
4. **RAG (Retrieval-Augmented Generation)** – Fetches relevant interview questions and guidelines from the knowledge base first, then generates tailored questions and answers, making the Interview Trainer Agent more accurate and trustworthy
5. **IBM watsonx.ai** – For model deployment, embeddings, and AI governance
6. **Vector Database (FAISS/Chroma)** – For RAG-based retrieval of interview questions, industry expectations, and HR guidelines
7. **Streamlit** – For the frontend, providing an interactive dashboard where users input their profile, upload a resume/JD, and view their personalized interview prep kit

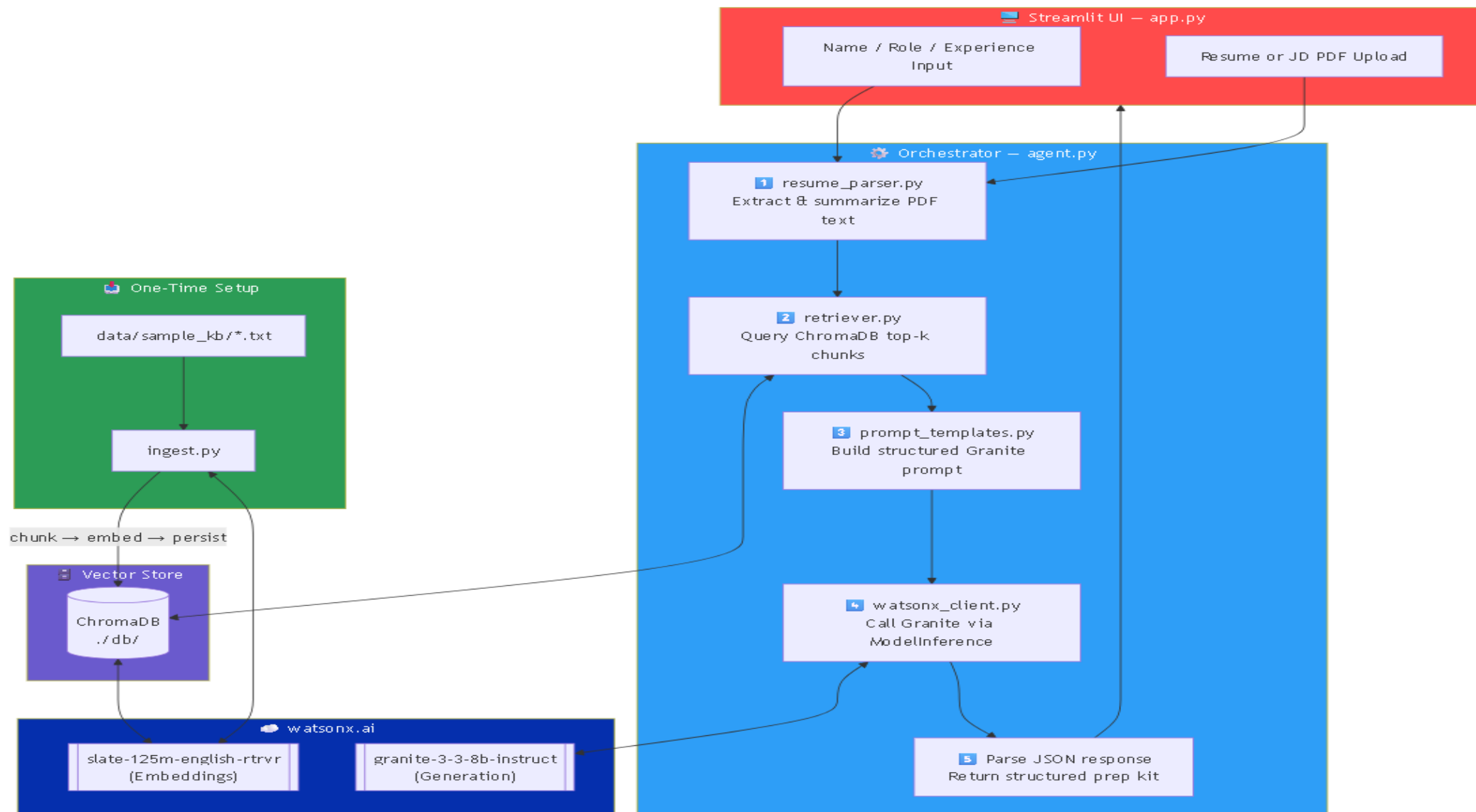
Key Modules Used -

1. **Frontend (Streamlit)** – Interactive dashboard where users enter their job title, experience level, and upload their resume or job description (PDF/text); built and deployed on Streamlit Community Cloud.
2. **Resume/JD Parser** – Extracts and summarizes text from the uploaded resume or job description to inform question personalization.
3. **Retriever Module (Chroma/FAISS)** – Retrieves the most relevant chunks from the knowledge base text files based on the user's role and experience level.
4. **IBM watsonx.ai Integration (Granite Model)** – Granite-3.3-8B-Instruct → Combines the user's profile and retrieved context to generate personalized interview questions and answers.
5. **Output Module (Streamlit Display)** – Displays AI-generated technical questions, behavioral/HR questions, model answers, and improvement tips.
6. **Prep Kit Export Module** – Allows users to download the generated interview prep kit as a Markdown file for offline practice.

System workflow -



Architecture blueprint



Interview Trainer Agent - Home Screen :

Interview Trainer Agent

Get a personalised interview prep kit — technical questions, behavioral STAR-format questions, model answers, and improvement tips — powered by IBM Granite on watsonx.ai.



Technical Questions

5 role-specific questions with model answer outlines



Behavioral / HR

3 STAR-format questions with structured answer frameworks



Improvement Tips

2 targeted tips per question to sharpen your delivery



Confidence Checklist

Personalised preparation actions before your interview



Download Kit

Export the full prep kit as a Markdown file

👉 Fill in your profile in the sidebar and click 🚀 Generate Prep Kit to get started.

Personalized Profile Input :

Your Profile

Full Name



Shraddha

 Target Role



Machine Learning Engineer



 Experience Level

Entry Level (0-2 years)



 Resume / Job Description



Shraddh...Resume.pdf
93.1KB



Shraddha_Bajpai_Resume.pdf uploaded

 Settings

Knowledge base chunks



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Deploy



Your personalised prep kit is ready!

Interview Prep Kit

Machine Learning Engineer · Entry Level (0-2 years) · Shraddha

↓ Download Markdown



Technical Questions

> Q1: How does linear regression handle non-linear relationships?

> Q2: What is the purpose of cross-validation in machine learning?

> Q3: Describe the differences between supervised, unsupervised, and reinforcement learning.

> Q4: How do you approach feature selection in a machine learning project?

> Q5: Explain the concept of bias-variance tradeoff in machine learning models.

Generated Prep Kit Overview :

✓ Shraddha_Bajpai_Resume.pdf uploaded

Settings

Knowledge base chunks



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Shraddha

Machine Learning Engineer

Entry Level (0-2 years)

🔥 Generate Prep Kit

> ⓘ How it works

Technical Questions

> Q1: How does linear regression handle non-linear relationships?

✓ Q2: What is the purpose of cross-validation in machine learning?

Model Answer

Cross-validation is a technique used to evaluate the performance of a machine learning model on unseen data. It helps prevent overfitting by training and testing the model on multiple subsets of the available dataset.

💡 Improvement Tips

💡 Explain k-fold cross-validation and its advantages over simple train-test splits.

💡 Discuss how cross-validation can help in hyperparameter tuning.

> Q3: Describe the differences between supervised, unsupervised, and reinforcement learning.

> Q4: How do you approach feature selection in a machine learning project?

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Technical Questions with Model Answers :

Created by [Admin](#) | [View Profile](#)

 **Your Profile**

Full Name 

Shraddha

Target Role 

Machine Learning Engineer 

Software Engineer

Data Analyst

Data Scientist

Product Manager

DevOps Engineer

Machine Learning Engineer

Other (specify below)

df uploaded

 **Settings**

Deploy 

> Q3: Describe the differences between supervised, unsupervised, and reinforcement learning.

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▼ Q4: How do you approach feature selection in a machine learning project?

Model Answer

Feature selection involves choosing the most relevant features from the dataset to improve model performance. This can be done using techniques like correlation analysis, recursive feature elimination, or mutual information scoring.

💡 Improvement Tips

💡 Talk about the importance of avoiding multicollinearity among selected features.

💡 Discuss how dimensionality reduction techniques like PCA can aid in feature selection.

> Q5: Explain the concept of bias-variance tradeoff in machine learning models.

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Behavioral / HR Questions — STAR Format

Behavioral HR Questions (STAR Format) :

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Machine Learning Engineer

Entry Level (0-2 years)

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💡 Behavioral / HR Questions — STAR Format

▼ HR1: Tell me about a project where you applied machine learning to solve a problem. What was your role, and what did you learn from the experience?

★ STAR Answer Outline

In my final year project, I worked on a team developing a predictive model for electric vehicle adoption. My role involved data preprocessing and feature engineering. Through this project, I learned the importance of data quality and the value of iterative model refinement based on feedback.

💡 Improvement Tips

💡 Highlight any challenges faced during the project and how they were overcome.

💡 Quantify the outcomes or improvements achieved through the application of machine learning.

> HR2: Can you describe a situation where you received feedback on your work? How did you respond to it, and what changes did you implement?

> HR3: Tell us about a time when you had to explain a complex technical concept to a non-technical audience. How did you approach this

Confidence Checklist :

✓ Shraddha_Bajpai_Resume.pdf uploaded

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Knowledge base chunks



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Shraddha

Machine Learning Engineer

Entry Level (0-2 years)

🔥 Generate Prep Kit

> ⓘ How it works

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✓ Confidence Checklist

- ✓ Review key concepts in machine learning, including supervised, unsupervised, and reinforcement learning paradigms.
- ✓ Practice implementing common algorithms like linear regression, decision trees, and clustering using Python libraries like scikit-learn.
- ✓ Familiarize yourself with data manipulation and visualization tools such as Pandas, NumPy, Matplotlib, and Seaborn.
- ✓ Develop a portfolio showcasing projects that demonstrate your skills in machine learning and data analysis.
- ✓ Prepare to discuss your experiences, challenges, and lessons learned from previous projects or internships.
- ✓ Research the company's current projects and initiatives in machine learning and artificial intelligence to show interest and enthusiasm.

Generated by Interview Trainer Agent · IBM Granite · watsonx.ai

Role of Agentic AI in Interview Trainer Agent

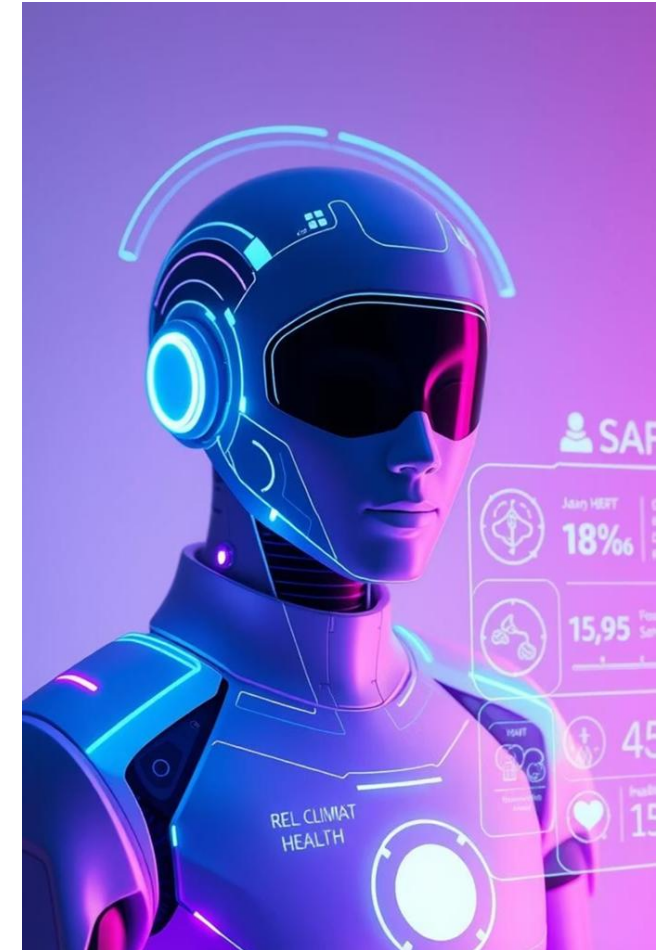
Agentic AI enables the Interview Trainer Agent to function as an intelligent, adaptive coach rather than a static question bank. Using IBM watsonx.ai and IBM Granite models, it coordinates specialized agents to deliver personalized interview preparation.

Each agent performs a specific role - such as retrieving role-specific interview questions, generating tailored answers, providing behavioral and HR coaching, and analyzing candidate responses - while working collaboratively.

Agentic AI also brings context awareness, understanding candidate details like job role, experience level, and resume background to generate tailored questions and answers. feedback. It supports real-time adaptation, continuously refining suggestions based on the candidate's specific profile and target position.

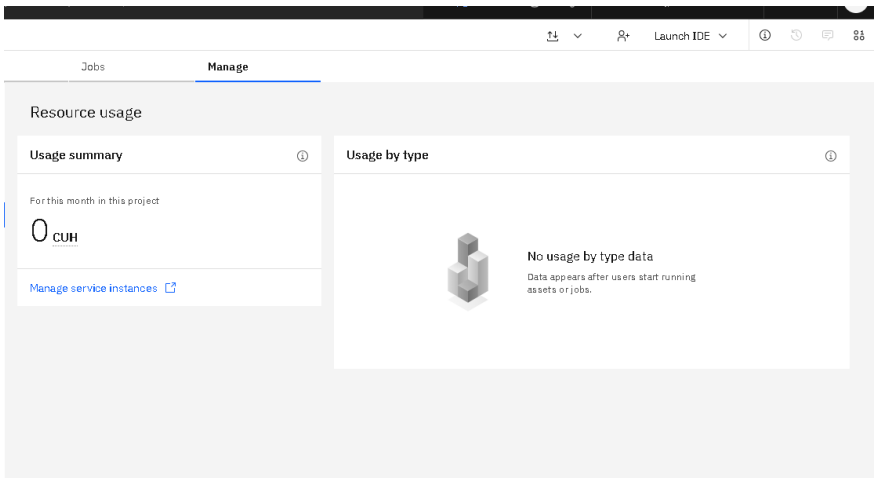
Additionally, agents communicate with each other, enabling better-grounded questions and more relevant improvement

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a smart digital interview coach that helps candidates prepare with confidence and perform at their best.

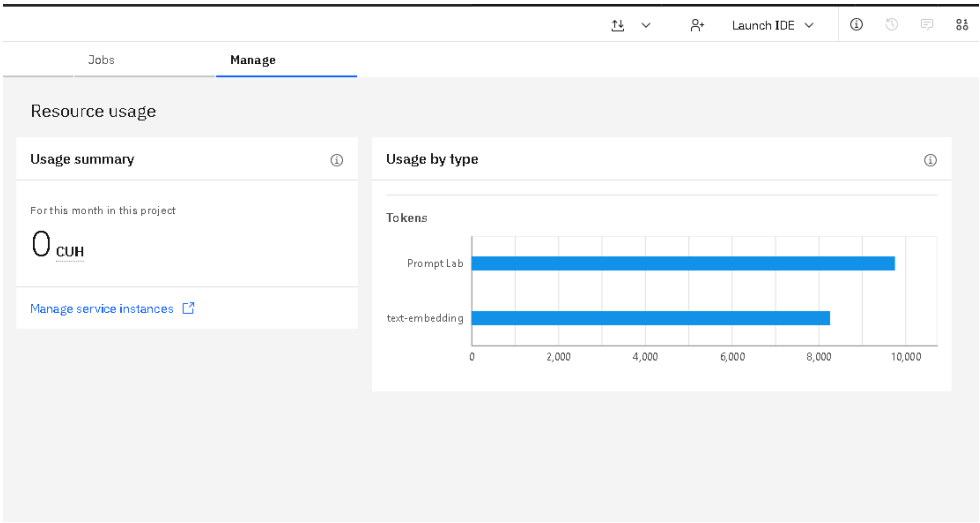


Token Usage

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed Interview Trainer Agent stands out by combining Agentic AI, RAG, and deep personalization into a single intelligent interview preparation system.

Multi-Agent Intelligence – Instead of a static question list, specialized agents collaborate to retrieve, generate, and evaluate interview content in a coordinated way.

RAG-Based Accuracy – Integrates real-time retrieval from a curated interview knowledge base with AI generation, reducing generic or irrelevant questions and ensuring grounded, trustworthy responses.

Hyper-Personalization – Tailors questions and answers based on the candidate's job role, experience level, and resume or job description.

Resume/JD-Aware Generation – Goes beyond generic interview prep by directly parsing the candidate's actual background to shape question relevance.

Comprehensive Skill Coverage – Combines technical questions with behavioral/HR questions in STAR format, addressing both hard skills and soft skills in one prep kit.

Actionable Feedback Loop – Provides model answers and targeted improvement tips alongside every question, not just a list of things to expect.

Scalable & Enterprise-Ready – Built on IBM watsonx.ai with IBM Granite models, ensuring reliability, governance, and room to scale beyond a hackathon prototype.

This makes the solution a smart, adaptive, and proactive digital interview coach, not just a static question repository.

- **Git Hub Link**

<https://github.com/Itsshraddha/Interview-Trainer-Agent.git>

- **Deployment Link :**

<https://interview-trainer-agentgit-h8n9bxx2jxumsvzchbneba.streamlit.app/>

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Future Scope

1. Integration with Job Portals & ATS Systems –

The system can be integrated with platforms like LinkedIn, Naukri, or company Applicant Tracking Systems to automatically pull real job descriptions and role requirements. This will enable the Interview Trainer Agent to generate even more accurate, up-to-date, and role-specific questions based on actual live job postings rather than static templates.

2. AI-Powered Mock Interview & Multilingual Support –

Enhancing the system with a voice-enabled mock interview mode - where users can speak their answers and receive real-time feedback on clarity, pacing, and content - along with support for multiple languages, will improve accessibility and realism. Users can practice naturally in their preferred language, making the solution more inclusive and suitable for diverse candidates across regions and linguistic backgrounds.

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Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Shraddha Bajpai

Has successfully satisfied the requirements for:

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Completion Certificate



This certificate is presented to
Shraddha Bajpai

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 30 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.